



Large-Language Models

Course AIAA 4220

Week 11- Lecture 21

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Lecture Overview

- ✓ **NLP Research Overview**
 - Word/Token Representation
 - Sentence Modeling
- ✓ **Representation Learning for LLM (also used for VLM/VLA)**
 - Pretraining, Finetuning, and Prompting
 - Post-training/Alignment
 - Reinforcement Learning from Human Feedback (RLHF)
- ✓ **Vision-Language-Models**
 - CLIP
 - LLaVA, QwenVL

- Technology to handle human language (usually text) using computers
- Aid **human-human communication** (e.g. machine translation)
- Aid **human-machine communication** (e.g. question answering, dialog)
- **Analyze/understand language**
 - Word Segmentation and Morphology
 - Syntactic Parsing
 - Semantic Parsing, etc.

A General Framework for NLP Systems

- Formally, **create a function to map an input X (language) into an output Y.**

Input X

Text

Text

Text

Text

Output Y

Text in Other Language

Response

Label

Linguistic Structure

Task

Translation

Dialog

Text Classification

Language Analysis

- To create such a system, we can use
 - Manual creation of rules
 - Machine learning from paired data $\langle X, Y \rangle$

Most of NLP tasks are Seq2Seq tasks

Example Task: Review Sentiment Analysis

- Given a review on a reviewing web site (X), decide whether its label (Y) is positive (1), negative (-1) or neutral (0)

I hate this movie →
positive
neutral
negative

I love this movie →
positive
neutral
negative

I saw this movie →
positive
neutral
negative

Example Task: Review Sentiment Analysis

- Some Difficult Cases
 - Negation

This one is not nearly as dreadful as expected.

positive

Serving Sara doesn't serve up a whole lot of laughs.

negative

- **Solution:** If a negation modifies a word, disregard it. Sometimes LLMs make mistake on complex negation as well

Example Task: Review Sentiment Analysis

- Some Difficult Cases
 - Metaphor, Analogy

Has all the depth of a wading pool.
negative



- **Solution?: ??** Ask LLM?



You

Is this movie review positive or negative: Has all the depth of a wading pool.



ChatGPT

This movie review is negative. Comparing something to "the depth of a wading pool" implies a lack of depth or substance, which is typically a criticism when it comes to narratives and storytelling.

Word Representations

- First, what is a word?
 - The things that are in the dictionary?
 - But how did the lexicographers decide what to put in the dictionary?
 - The things between spaces and punctuation?
- So to get words, we need to do tokenization
 - Which is the process of getting the smallest units for NLP systems to process
 - Can be non-trivial for many languages

Word Representations

- Linguistic Morphology
 - **Morphology is the study of the internal structure of words**
 - **Derivational morphology.** How new words are created from existing words.
 - [grace]
 - [[grace]ful]
 - [un[grace]ful]]
 - **Inflectional morphology.** How features relevant to the syntactic context of a word are marked on that word.
 - This student walks.
 - These students walk.
 - These students walked.
 - **Compounding.** Creating new words by combining existing words
 - With or without spaces: surfboard, golf ball, blackboard

Word Representations

- So a good tokenization method can get morphemes extracted
- Tokenization approaches
 - Unsupervised subword segmentation
 - Instead of
 - white-space segmentation
 - single-character segmentation
 - **Use the data** to tell us how to tokenize
- **Subword tokenization approaches**
 - Three common algorithms:
 - **Byte-Pair Encoding (BPE)** (Sennrich et al., 2016) (used in transformers, CLIP, etc.)
 - Unigram language modeling tokenization (Kudo, 2018)
 - WordPiece (Schuster and Nakajima, 2012)
 - All have 2 parts:
 - A **token learner** that takes a raw training corpus and induces a vocabulary (a set of tokens)
 - A token **segmenter** that takes a raw test sentence and tokenizes it according to that vocabulary

Word Representations

- Byte Pair Encoding (BPE)
 - Properties of BPE tokens
 - Usually include frequent words
 - And frequent subwords
 - Which are often morphemes like -est or -er or un- or -ful
 - Used in most LLMs / VLMs / VLAs

Sentence Modeling

Sequence Label Prediction



Masterful!

By Antony Witheyman - January 12, 2006

Ideal for anyone with an interest in
disguises who likes to see the subject
tackled in a humourous manner.

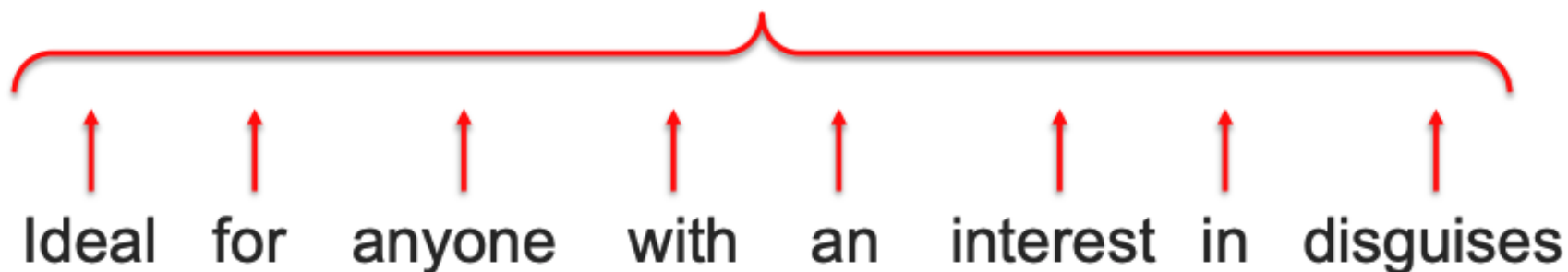
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Prediction



Sentiment ?
(positive or negative)

Sentiment label?



Sentence Modeling

Sequence Prediction



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0 of 4 people found this review helpful

Prediction



Part-of-speech ?
(noun, verb,...)

POS?

POS?

POS?

POS?

POS?

POS?

POS?

POS?



Ideal

for

anyone

with

an

interest

in

disguises

Sentence Modeling

Language Model



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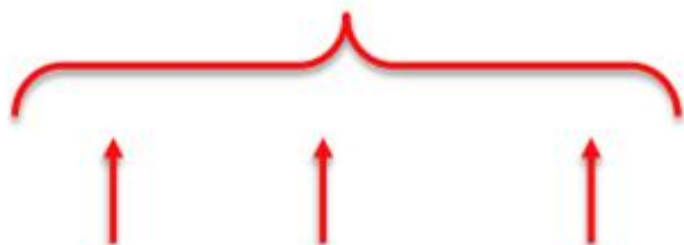
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Prediction



Language Model

Next word?



Ideal for anyone

with an interest in disguises

Sentence Modeling

Language Model

Embedding

[0,1;
0,0004;
....;
0.09;
0,05]

Generation



Ideal for anyone with an interest in
disguises who likes to see the subject
tackled in a humourous manner.

Example: Image captioning



[0,1;
0,0004;
....;
0.09;
0,05]



The man at bat readies to swing at the
pitch while the umpire looks on.

Sentence Modeling

Challenges in Sequence Modeling



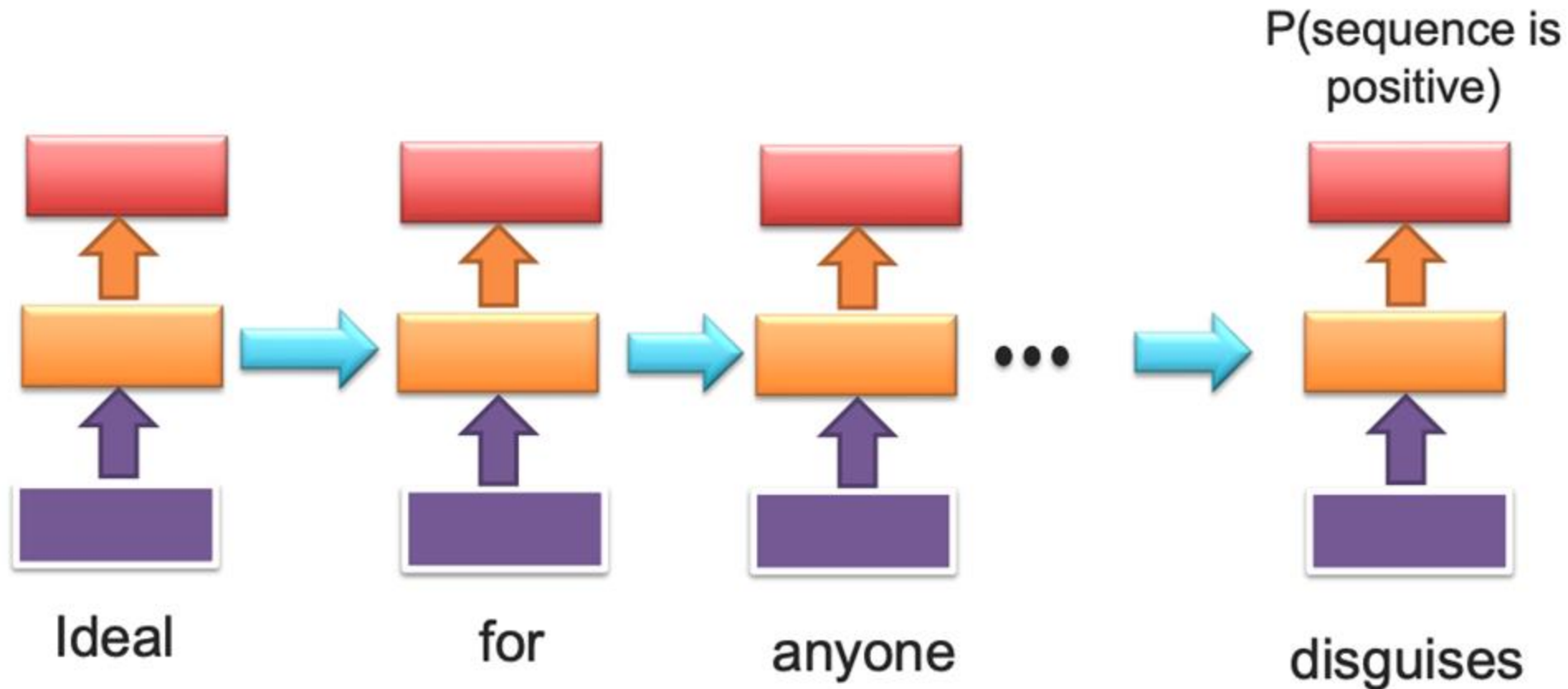
- Part-of-speech ?
(noun, verb,...)
- Sentiment ?
(positive or negative)
- Language Model
- Sequence representation

- Main Challenges:
 - Sequences of variable lengths (e.g., sentences)
 - Keep the number of parameters at a minimum
 - Take advantage of possible redundancy

Sequence Modeling Methods

RNN (LSTM)

- RNN for Sequence Label Prediction

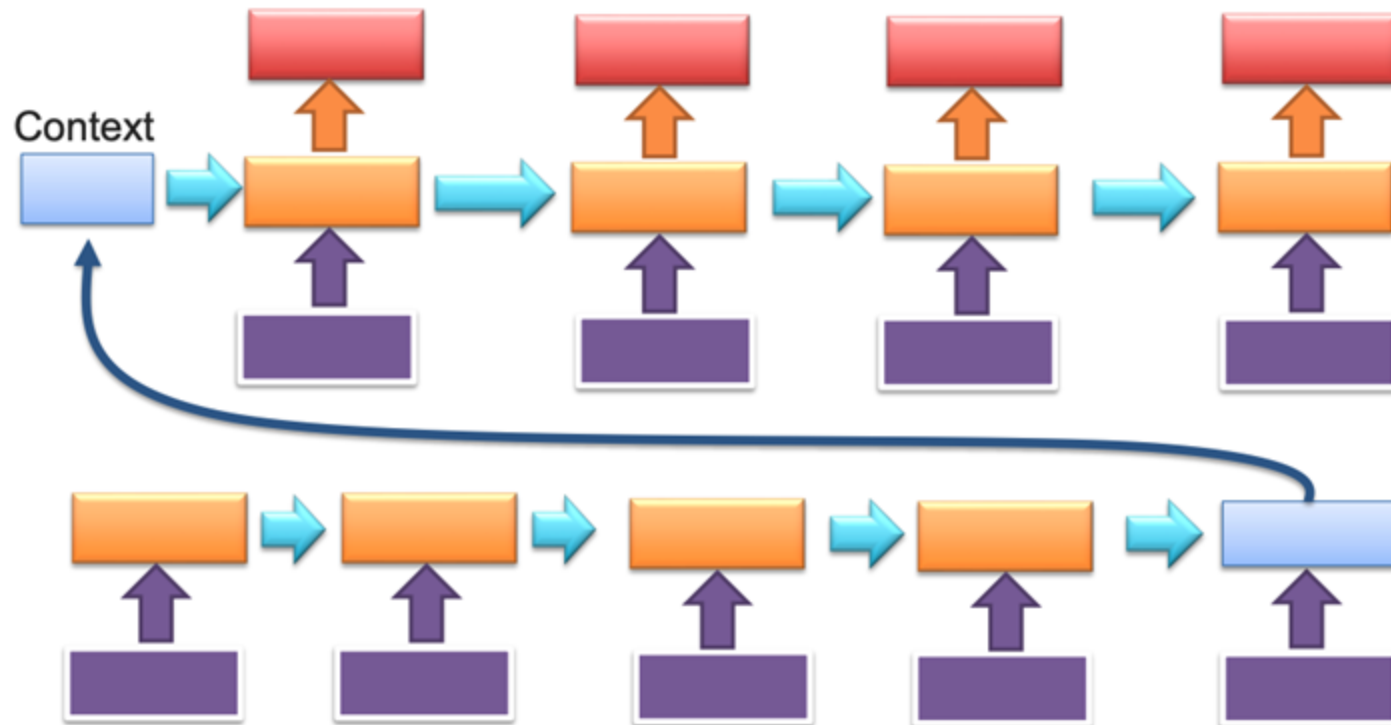


Sequence Modeling Methods

RNN (LSTM)

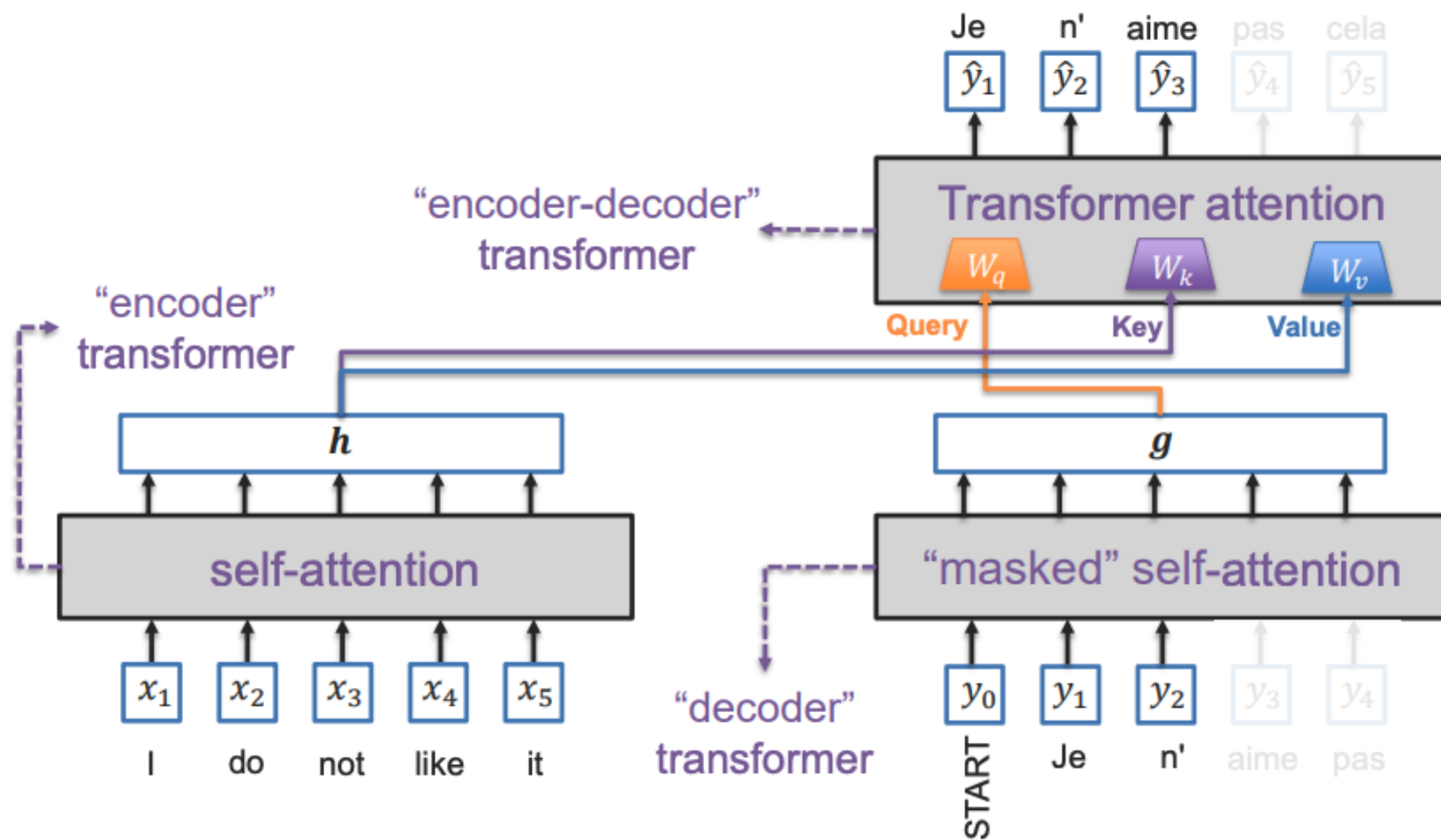
- RNN for Machine Translation - Encoder-Decoder Architecture

Le chien sur la plage → The dog on the beach



Sequence Modeling Methods

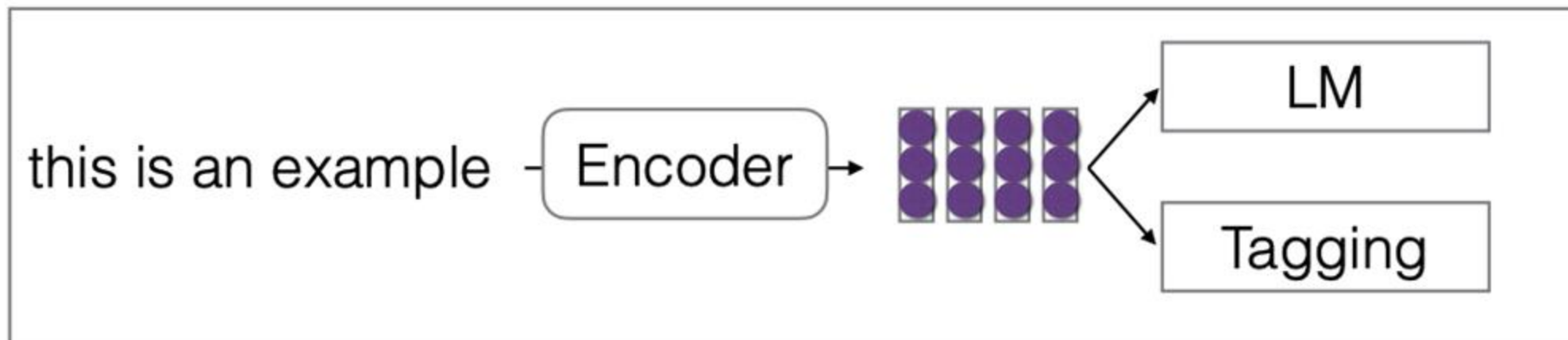
Transformers



Representation Learning for LLM

Standard Multi-task Learning in NLP

- Train representations to do well on multiple tasks at once

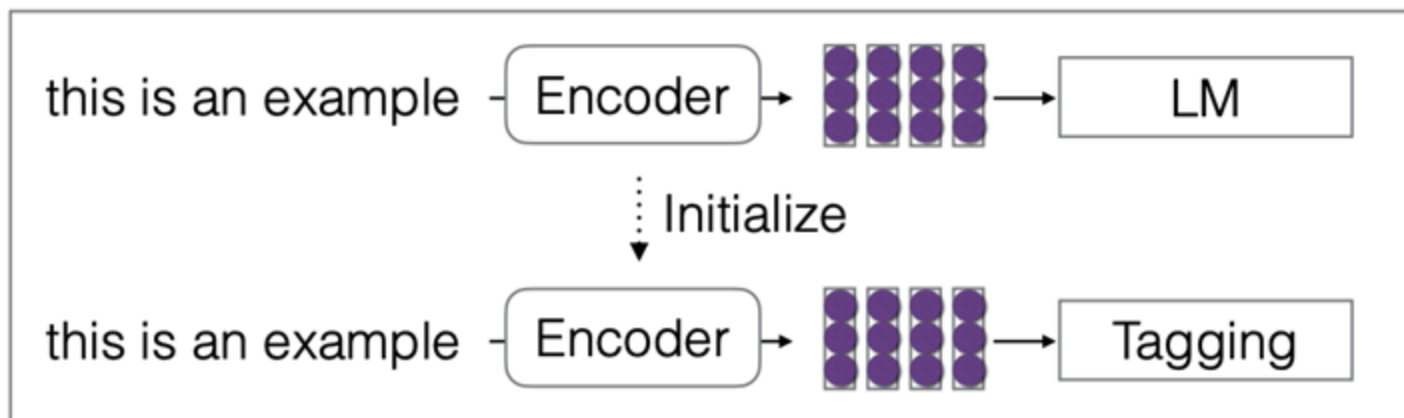


- Often as simple as randomly choosing minibatch from one of multiple tasks

Representation Learning for LLM

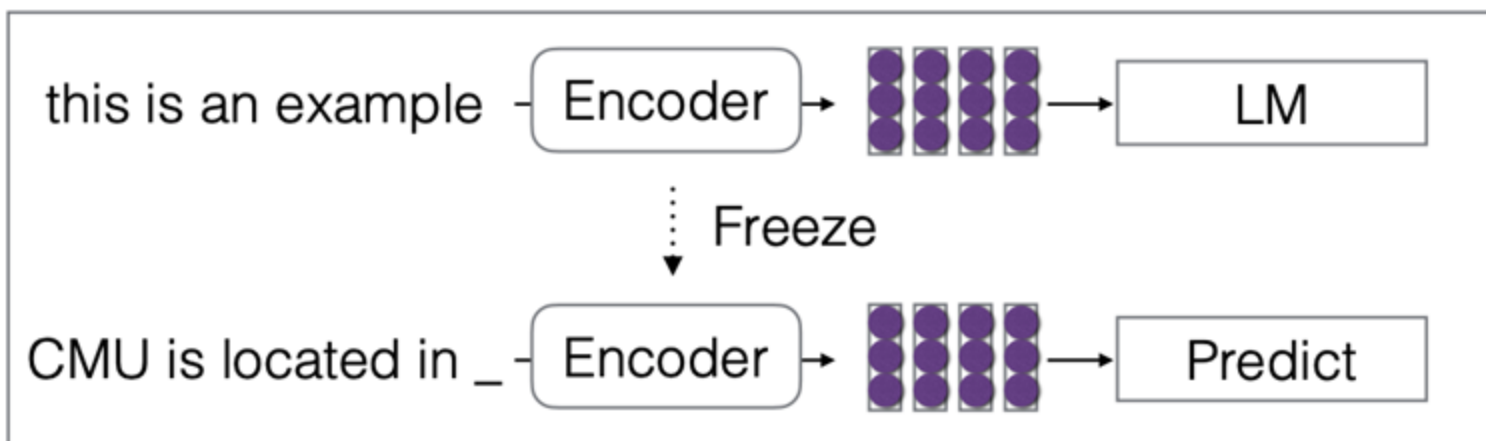
Pre-train and Fine-Tune

- First train on one task, then train on another



Prompting

Train on LM task, make predictions in textualized tasks



Representation Learning for LLM

Pre-trained LMs

- Many pre-trained LMs have names like BERT, RoBERTa, GPT-3, PaLM
- These often refer to a combination of
 - Model: The underlying neural network architecture
 - Training objective: What objective is used to pretrain
 - Data: What data the authors chose to use to train the model

Model:

- Usually **Transformer**, although the details vary
- **Size** is an all-important parameter, bigger is usually more performant
- Model details sometimes vary (or are underspecified)

Data is extremely important, common sources

Books corpus - a large corpus of books

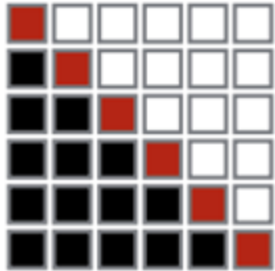
Wikipedia

Common crawl - data from the whole internet

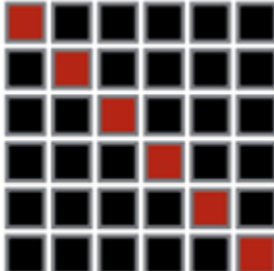
Representation Learning for LLM

Which Objective?

- Two most common varieties
 - **Auto-regressive language modeling** -> used more for prompting/text generation

$$P(X) = \prod_{i=1}^{|X|} P(x_i | x_1, \dots, x_{i-1})$$


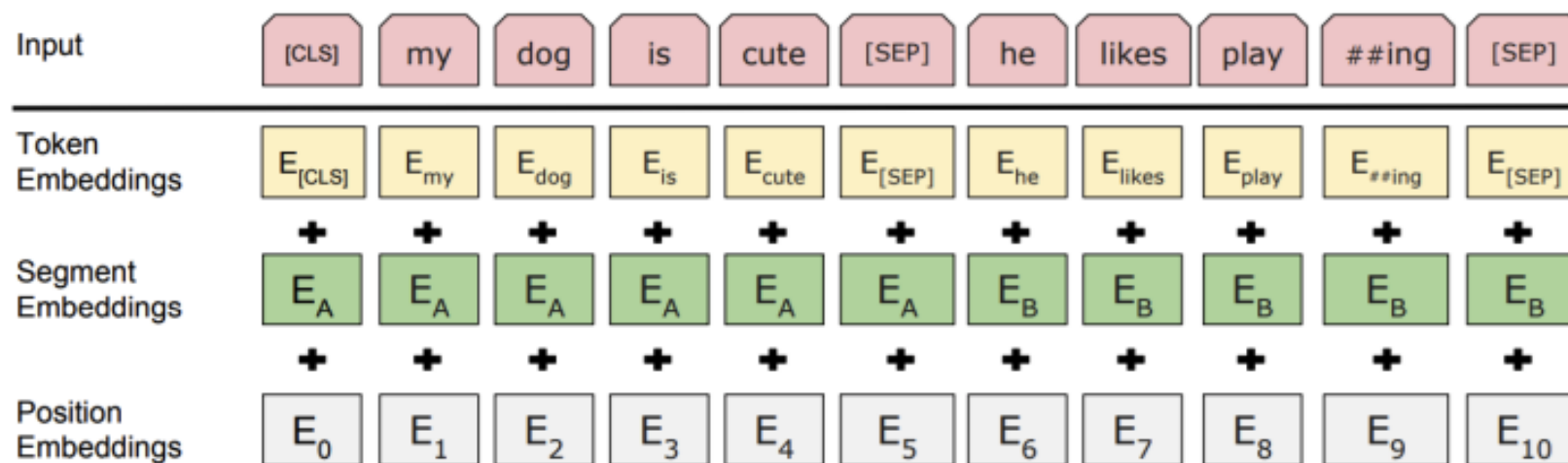
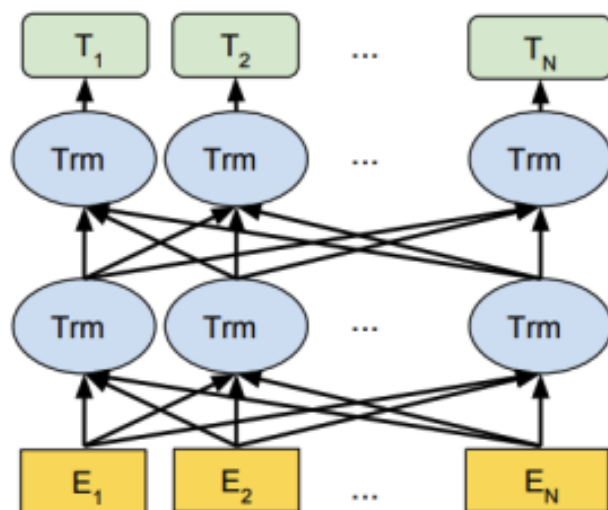
- **Masked language modeling** -> used more for pretrain + fine-tune

$$P(X) \neq \prod_{i=1}^{|X|} P(x_i | x_{\neq i})$$


Representation Learning through LMs

- **Masked Language Modeling (BERT)**

- Model: Multi-layer self-attention. Input sentence or pair, w/ [CLS] token, subword representation
- Objective: Masked word prediction + next sentence prediction
- Data: BooksCorpus + English Wikipedia



Representation Learning through LMs

- **Masked Language Modeling (BERT)**
 - Task 1. predict a masked word
 - Substitute input word with [MASK]
 - Task 2. classify two sentences as consecutive or not:
 - 50% of training data (from OpenBooks) is "consecutive"

Input = [CLS] the man [MASK] to the store [SEP]
penguin [MASK] are flight ##less birds [SEP]

Label = NotNext

Input = [CLS] the man went to [MASK] store [SEP]
he bought a gallon [MASK] milk [SEP]

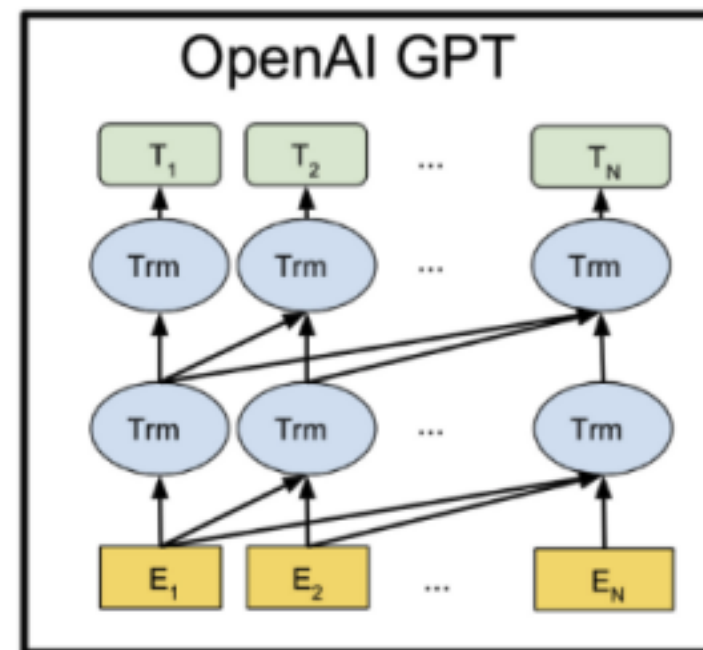
Label = IsNext

Representation Learning through LMs

Auto-regressive LMs for Generation/Prompting

- **GPT-2**

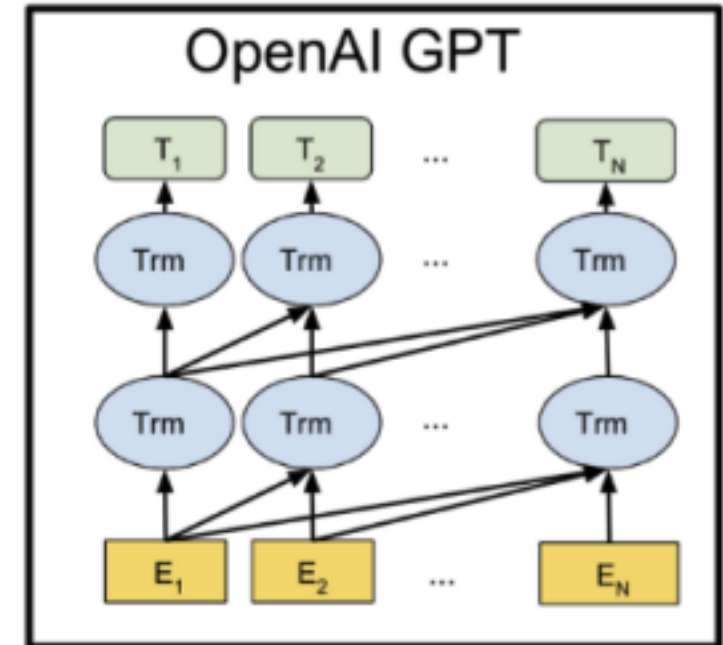
- Model: Left-to-right transformer (1.5B)
- Training Objective: Standard language modeling
- Data: WebText (millions of web pages)
 - Scrape links posted on Reddit w/ at least 3 upvotes (rough proxy of human quality)
- Results: Impressive results in generation of longform text, and zero shot task completion
- Available open source, easy to use



Representation Learning through LMs

Auto-regressive LMs for Generation/Prompting

- **GPT-3**
 - Model: Left-to-right transformer (175B) (100x vs GPT-2)
 - Training Objective: Standard language modeling
 - Data: CommonCrawl (1T words)
 - Results: Further impressive results in generation of long-form text, and zero shot task completion



Representation Learning through LMs

Scaling Laws (Kaplan et al. 2020)

- Language models exhibit slow-fast-slow learning pattern
- Larger models improve in fewer steps

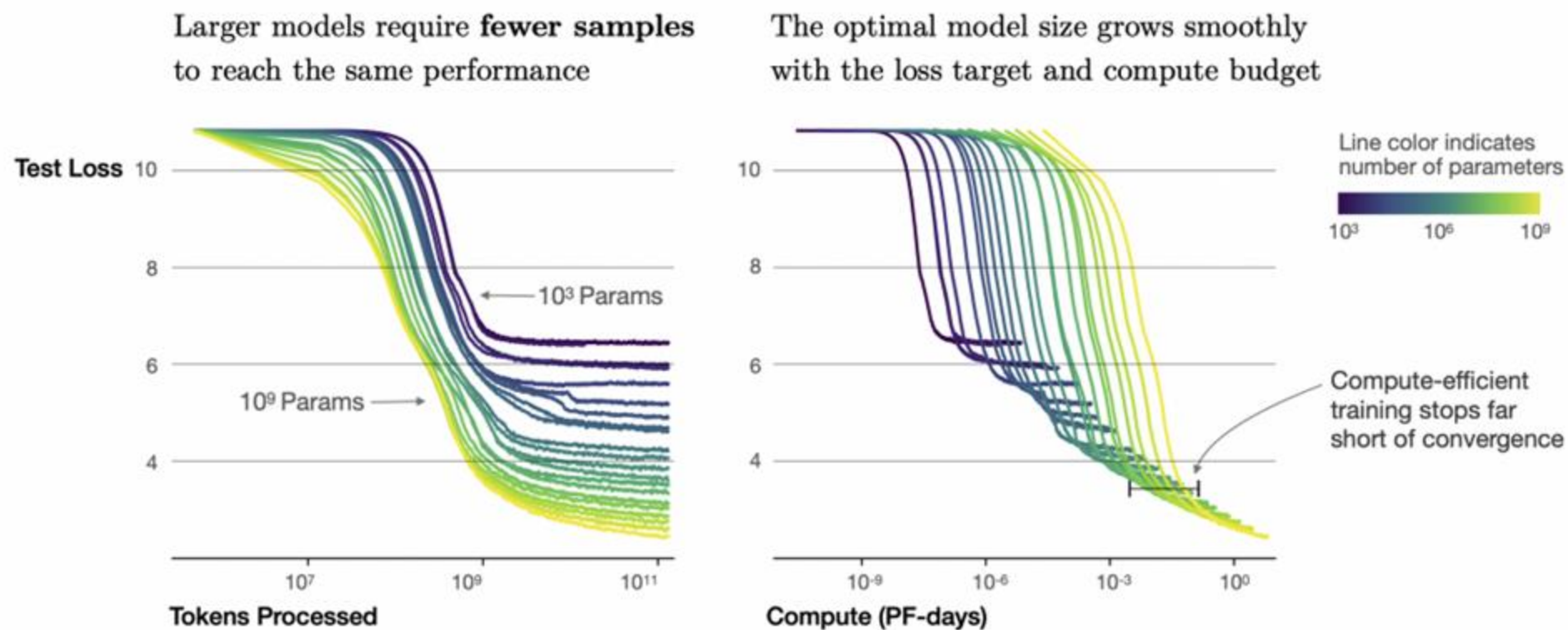


Figure 2 We show a series of language model training runs, with models ranging in size from 10^3 to 10^9 parameters (excluding embeddings).

Four Paradigms of NLP Progress

Feature Engineering

- Paradigm: Fully Supervised Learning (Non-neural Network)
- Time Period: Most popular through 2015
- Characteristics:
 - Non-neural machine learning models mainly used
 - Require manually defined feature extraction
- Representative Work:
 - Manual features -> linear or kernelized support vector machine (SVM)
 - Manual features -> conditional random fields (CRF)

Architecture Engineering

- Paradigm: Fully Supervised Learning (Neural Networks)
- Time Period: About 2013-2018
- Characteristics:
 - Rely on neural networks
 - Do not need to manually define features, but should modify the network structure (e.g.: LSTM v.s CNN)
 - Sometimes used pre-training of LMs, but often only for shallow features such as embeddings
- Representative Work
 - CNN for Text Classification

Four Paradigms of NLP Progress

Objective Engineering

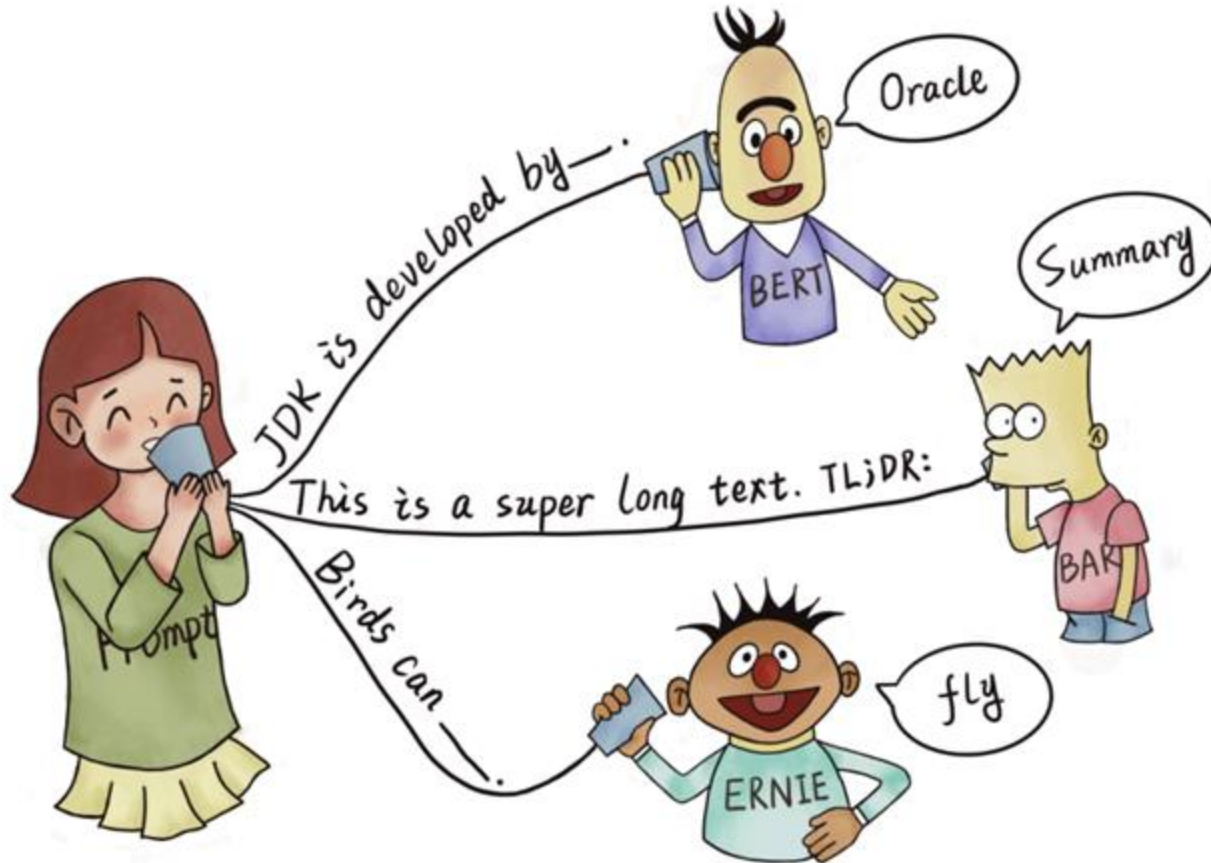
- Paradigm: Pre-train, Fine-tune
- Time Period: 2017-Now
- Characteristics:
 - Pre-trained LMs (PLMs) used as initialization of full model - both shallow and deep features
 - Less work on architecture design, but engineer objective functions
- Typical Work:
 - BERT → Fine Tuning

Prompt Engineering

- Paradigm: Pre-train, Prompt, Predict
- Date: 2019-Now
- Characteristic:
 - NLP tasks are modeled entirely by relying on LMs
 - The tasks of shallow and deep feature extraction, and prediction of the data are all given to the LM
 - Engineering of prompts is required
- Representative Work:
 - GPT 3

Prompting

- Encouraging a pre-trained model to make particular predictions by providing a "prompt" specifying the task to be done



The general workflow of Prompting:

- Prompt Addition
- Answer Prediction
- Answer-Label Mapping

Prompting

1. Prompt Addition

- Prompt Addition: Given **input x**, we transform it into prompt **x'** through two steps:
 - Define a template with two slots, one for input [x], and one for the **answer/output [z]**
 - Fill in the input slot [x]
- Example: Sentiment classification
 - z will tell us positive/negative

Input: x = "I love this movie"



Template: [x] Overall, it was a [z] movie

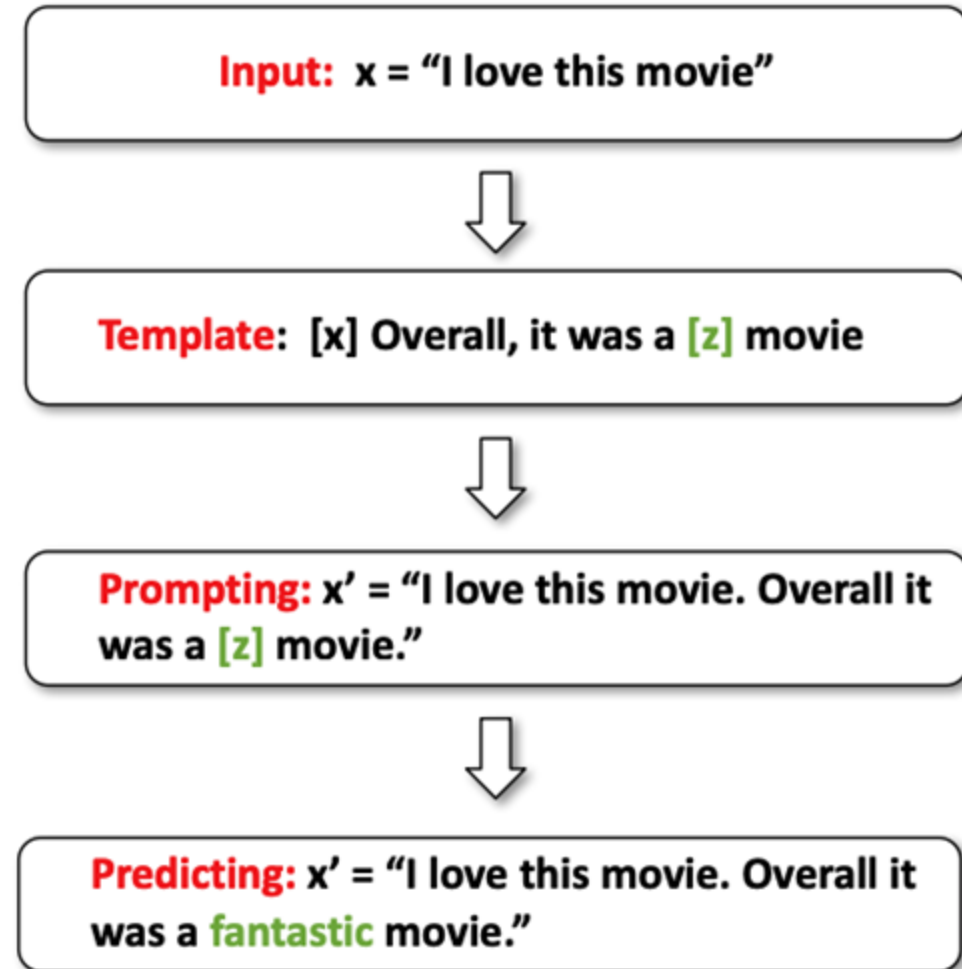


Prompting: x' = "I love this movie. Overall it was a [z] movie."

Prompting

2. Answer Prediction

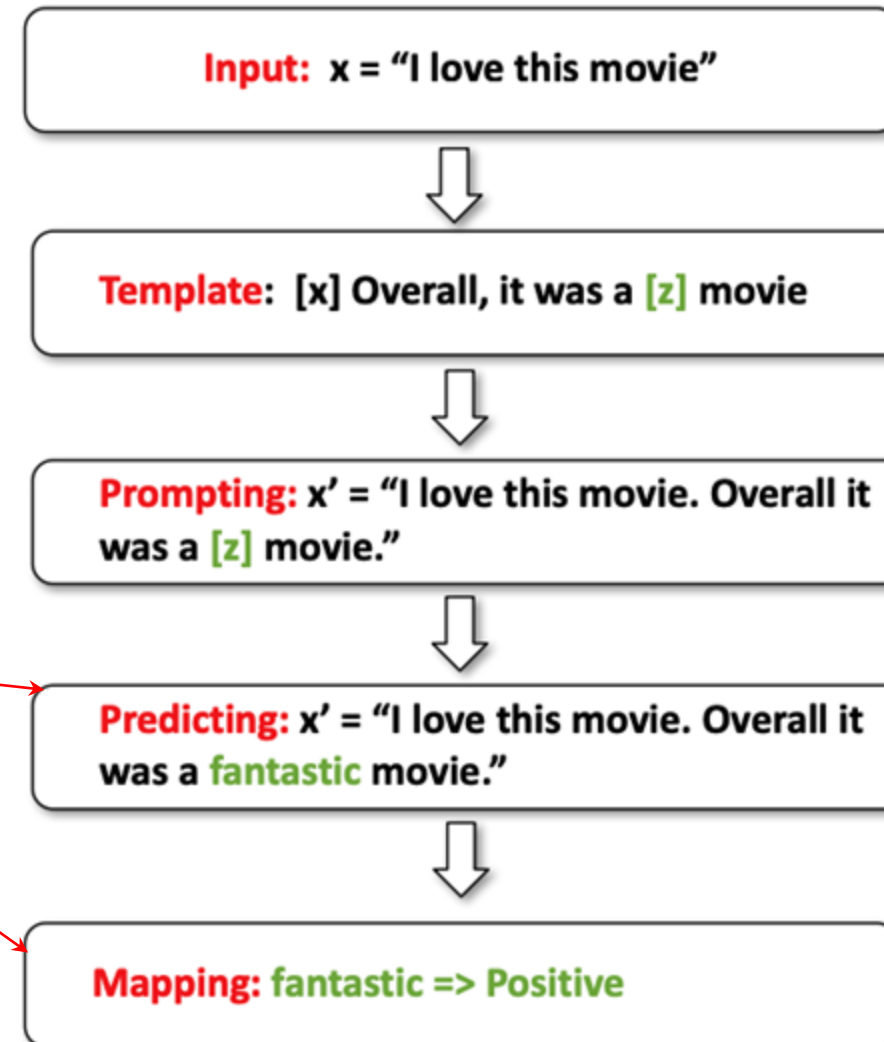
- Given this prompt, predict the answer [z]
 - Fill in [z]



Prompting

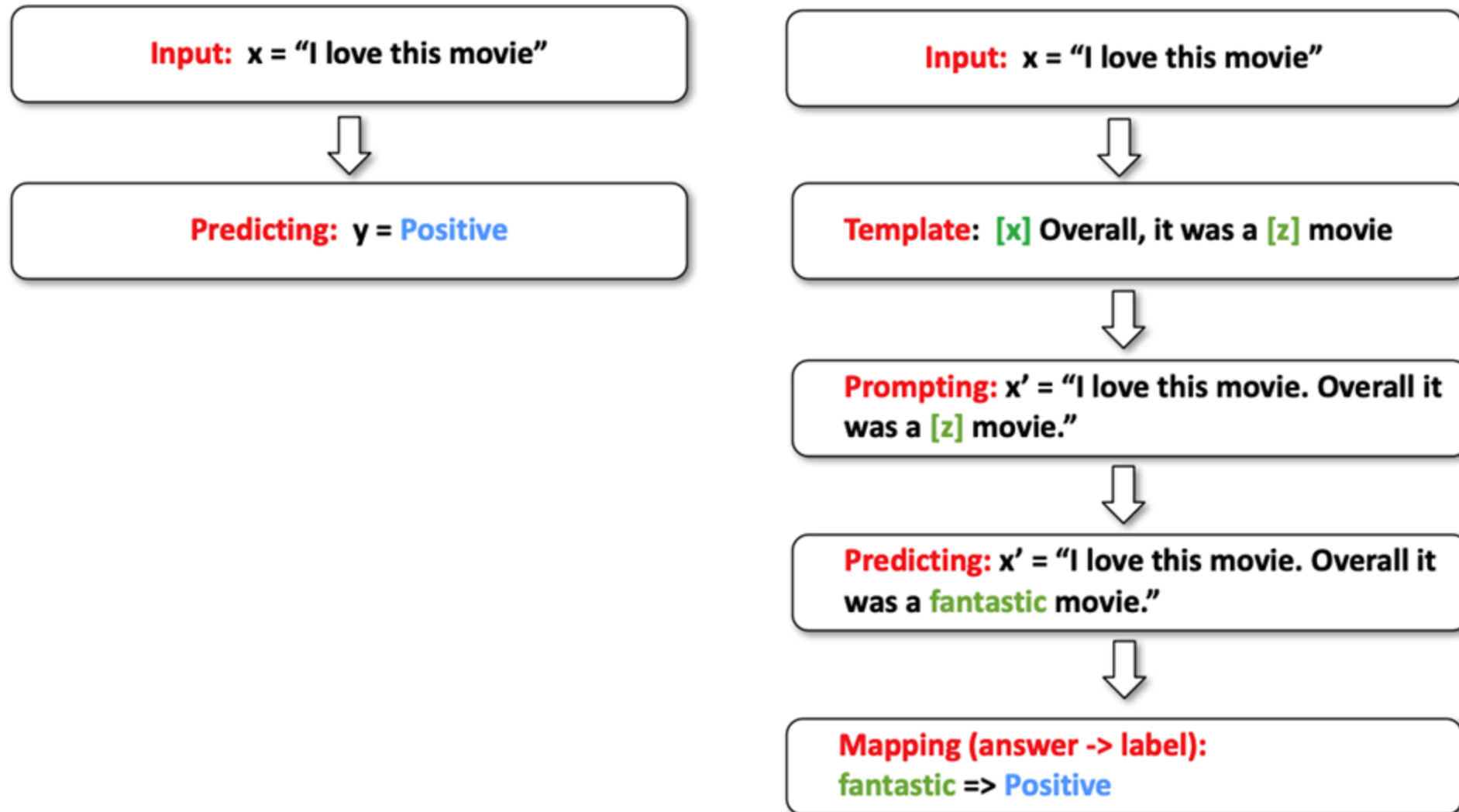
3. Answer-Label Mapping

- Given an answer, map it into a class label
 - Since language model will output a **distribution over all words**, and the word might not be exactly our target output
 - Example: Sentiment classification
 - Target output: positive/negative



Prompting

Traditional Formulation V.S Prompt Formulation

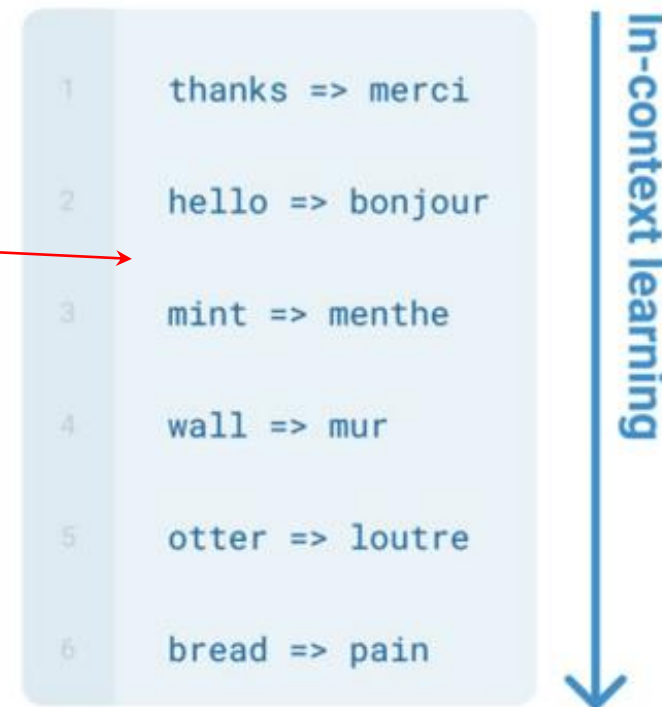


Emergent abilities of LLMs

GPT-3

- **Emergent few-shot learning with prompting**
 - Specify a task by simply prepending examples of the task before your example
 - Also called **in-context learning**, to stress that **no gradient updates** are performed when learning a new task (there is a separate literature on few-shot learning with gradient updates)

Use “=>” to represent
a translation task



Emergent abilities of LLMs

- **Limits of prompting for harder tasks?**

- Some tasks seem too hard for even large LMs to learn through prompting alone.
- Especially tasks involving **richer, multi-step reasoning**.
 - (Humans struggle at these tasks too!)

19583 + 29534 = 49117
98394 + 49384 = 147778
29382 + 12347 = 41729
93847 + 39299 = ?

Solution: change the prompt!

Chain-of-thought prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. 

So here one-shot prompting fails.

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

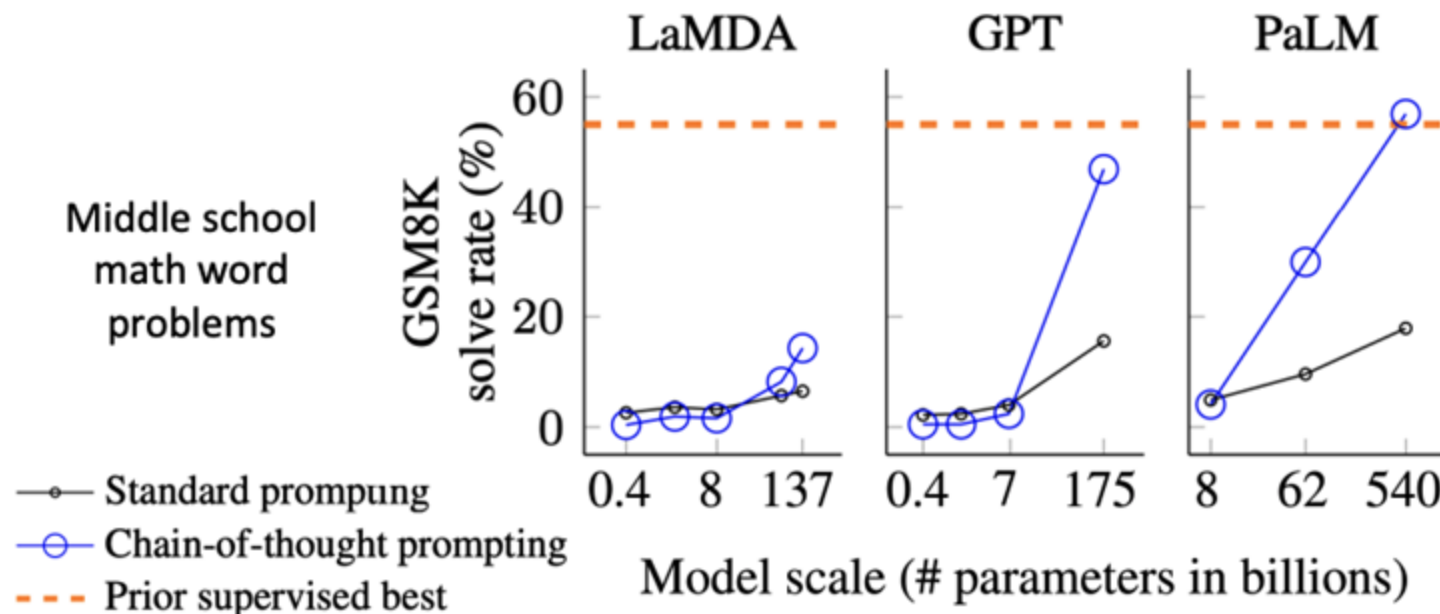
Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. 

Chain-of-thought one-shot prompting:
providing the reasoning process works!

Chain-of-thought prompting

- **Chain-of-Thought (CoT) Prompting** is a technique used to improve the performance of large language models (LLMs) on complex reasoning tasks, such as arithmetic, commonsense, and symbolic reasoning.
- Instead of just asking for a final answer, **you prompt the model to generate a series of intermediate reasoning steps that lead to the final answer.** This mimics a human's step-by-step problem-solving process.





Thanks for your attention!

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